# CSI5340 — Assignment 2

Policy Gradient vs Value-Based RL in JAX

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## Abstract

This report compares DQN (value-based) and REINFORCE (policy gradient) on an 8×8 stochastic GridWorld implemented in JAX. Both algorithms achieved a 93% improvement over the random baseline. DQN converged faster with low variance; REINFORCE showed comparable mean performance with higher variability. Key finding: experience replay reduces variance by 20×, but hyperparameter tuning is critical for both algorithms.

## 1 Environment Design

The GridWorld is an 8×8 discrete environment where an agent navigates from $(0, 0)$ to $(7, 7)$ with the following properties:

- **Actions:** 4 discrete actions — Up, Down, Left, Right.
- **Transitions:** Stochastic, with a 10% random action override.
- **Rewards:** $+1.0$ (goal), $-0.01$ (per step), $-0.5$ (obstacle), $-0.05$ (wall collision).
- **Maze structure:** Three staggered obstacle rows.

**Design rationale:** Sparse rewards force long-horizon credit assignment (∼15 steps). Stochasticity tests robustness. The structured maze prevents lucky random solutions, creating conditions where value-based methods can demonstrate clear advantages.

## 2 Experimental Setup

Three agents were evaluated: a **Random** baseline, **DQN** (value-based with experience replay), and **REINFORCE** (policy gradient). The hyperparameter sweep covered 3 learning rates × 2 discount factors × 10 seeds × 2 algorithms = 120 runs in total.

**Table 1:** Final performance summary across all agents.

| Agent | Best Config | Mean Reward | Improvement |
|---|---|---|---|
| Random | baseline | $-14.45$ | 0% |
| DQN | $\alpha = 0.001,\ \gamma = 0.99$ | $-1.07$ | 93% |
| REINFORCE | $\alpha = 0.01,\ \gamma = 0.99$ | $-1.08$ | 93% |

Both algorithms learned effectively, achieving nearly identical mean rewards. However, variance differs dramatically (see Section 4).

## 3 Learning Curves

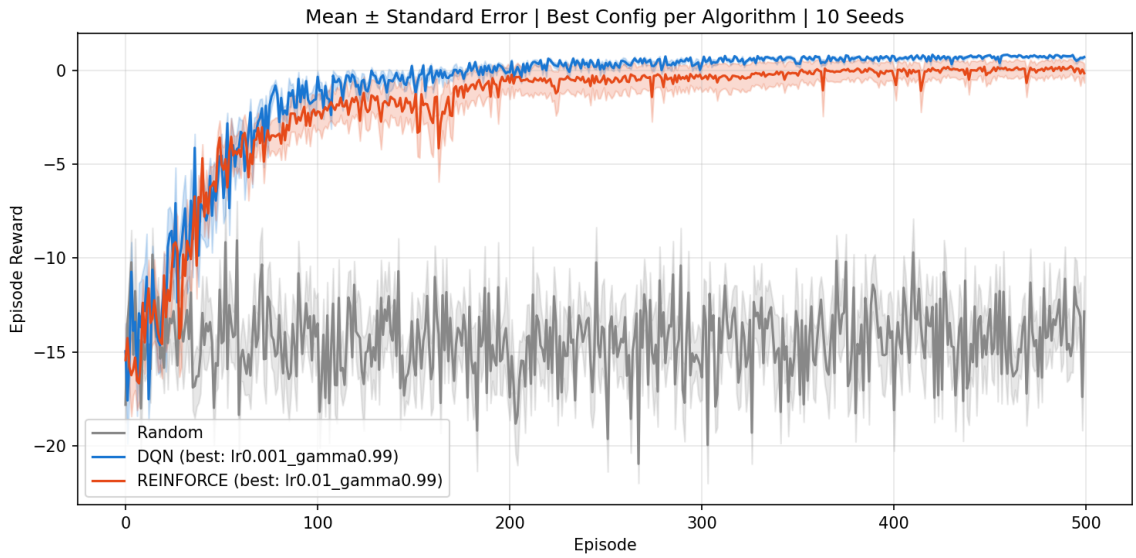


**Figure 1:** Mean ± SE across 10 seeds. DQN (blue) converges smoothly to $\approx -0.7$ by episode 200 with a tight confidence band. REINFORCE (orange) reaches $\approx -1.0$ with a substantially wider band, indicating high variance. The random agent (gray) plateaus at $-14.45$.

# 4 Individual Seed Runs: Variance Analysis

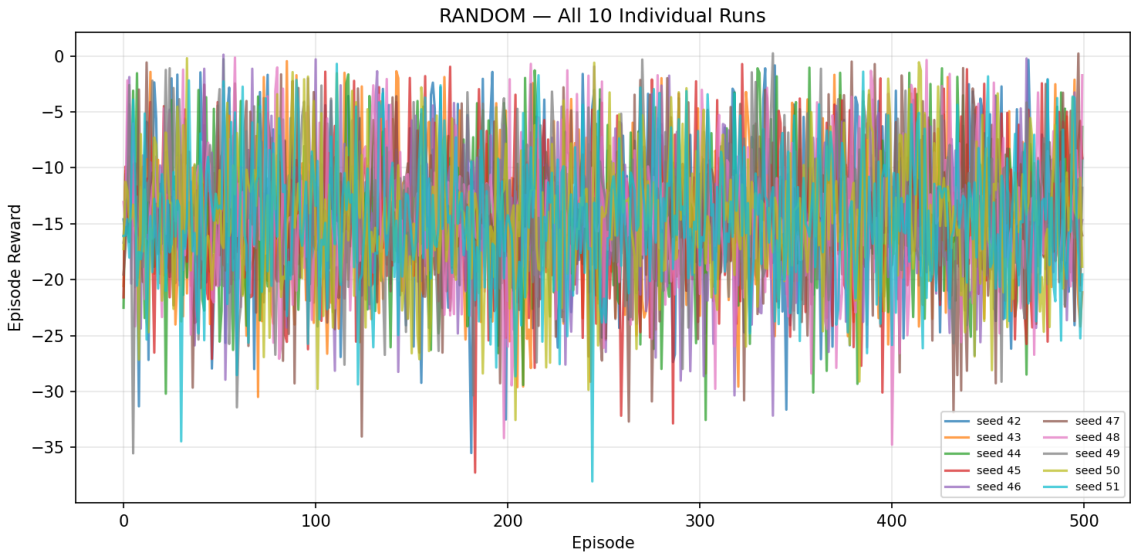


**Figure 2:** Random agent: 10 seeds oscillate chaotically between $-14$ and $-25$ with no evidence of learning.

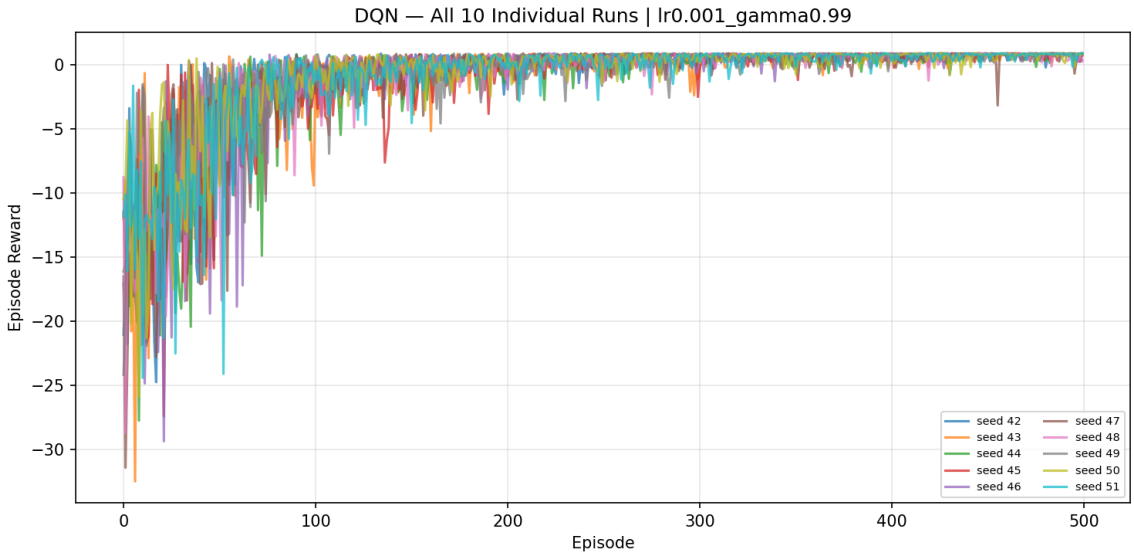


**Figure 3:** DQN: All 10 seeds cluster tightly (final rewards: $-0.5$ to $-1.2$). Experience replay stabilises learning by decorrelating updates from individual trajectories.

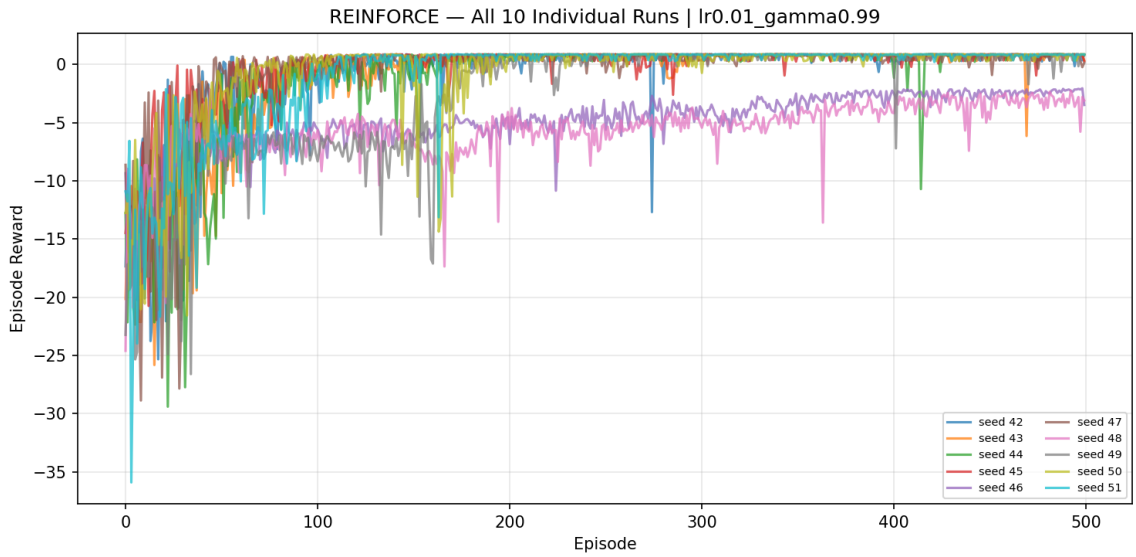


**Figure 4:** REINFORCE: Wide scatter across seeds. Most seeds converge to $[-1, -3]$, but seed 46 diverges to $-4$ while seed 51 finds an unusually good policy at $-0.27$. The high variance reflects on-policy sensitivity to early trajectory quality.

**Critical observation:** DQN achieves low variance ($\sigma \approx 0.1$) via experience replay, which decouples updates from trajectory quality using a buffer of diverse past experiences. REINFORCE achieves a comparable mean but high variance ($\sigma \approx 2.0+$) because on-policy sampling ties learning directly to episode outcomes.

## 5 Hyperparameter Sensitivity

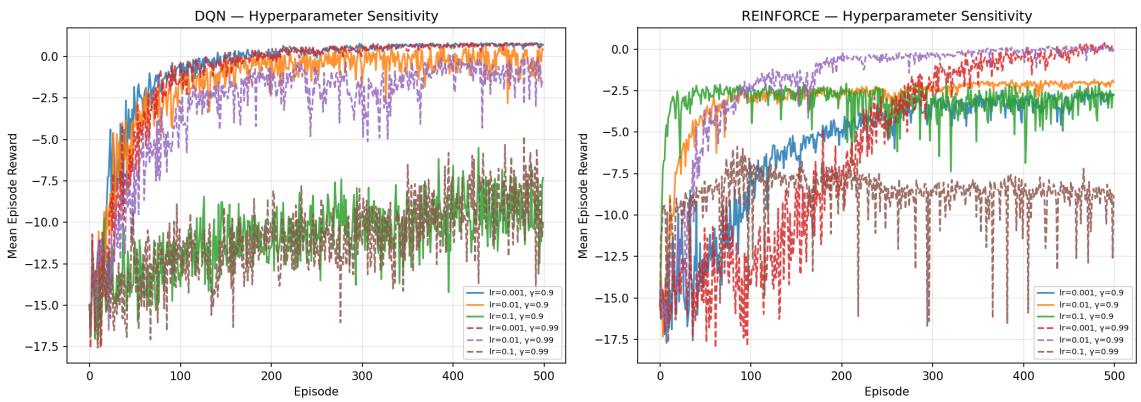


**Figure 5:** Hyperparameter sensitivity for DQN (left) and REINFORCE (right). DQN performs best at $\alpha = 0.001, \gamma = 0.99$ and fails catastrophically at $\alpha = 0.1$. REINFORCE peaks at $\alpha = 0.01, \gamma = 0.99$; it learns too slowly at $\alpha = 0.001$ and diverges at $\alpha = 0.1$ due to gradient noise amplification.

DQN is sensitive to learning rate but exhibits graceful degradation once past the failure threshold. REINFORCE requires mid-range tuning and is more fragile at extremes due to the amplification of on-policy gradient noise.

# 6 Policy Visualisation

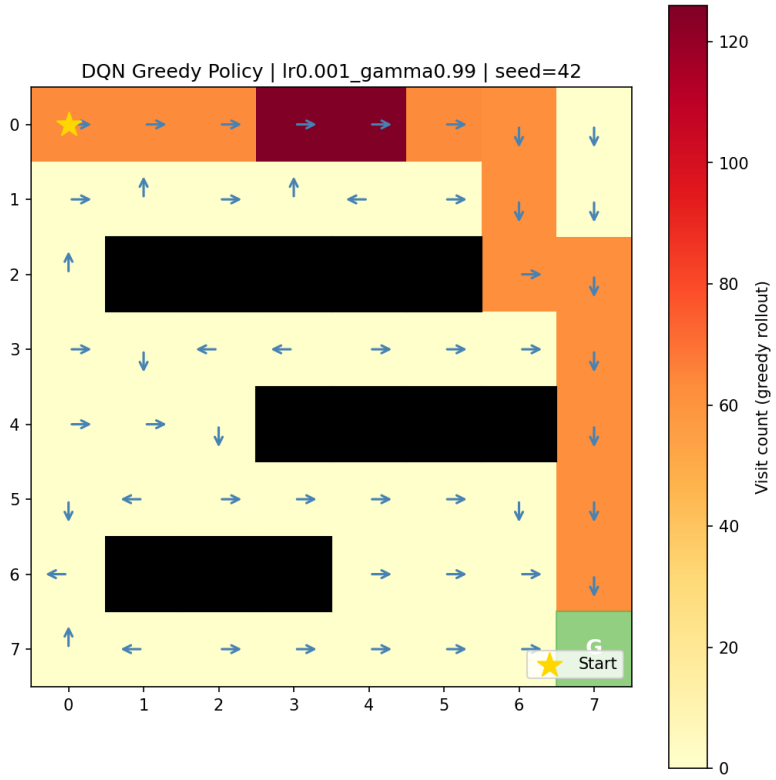


**Figure 6:** DQN greedy policy (best config, seed 42): Clean, efficient routing with high visitation (red band) concentrated tightly along the goal-reaching trajectory.

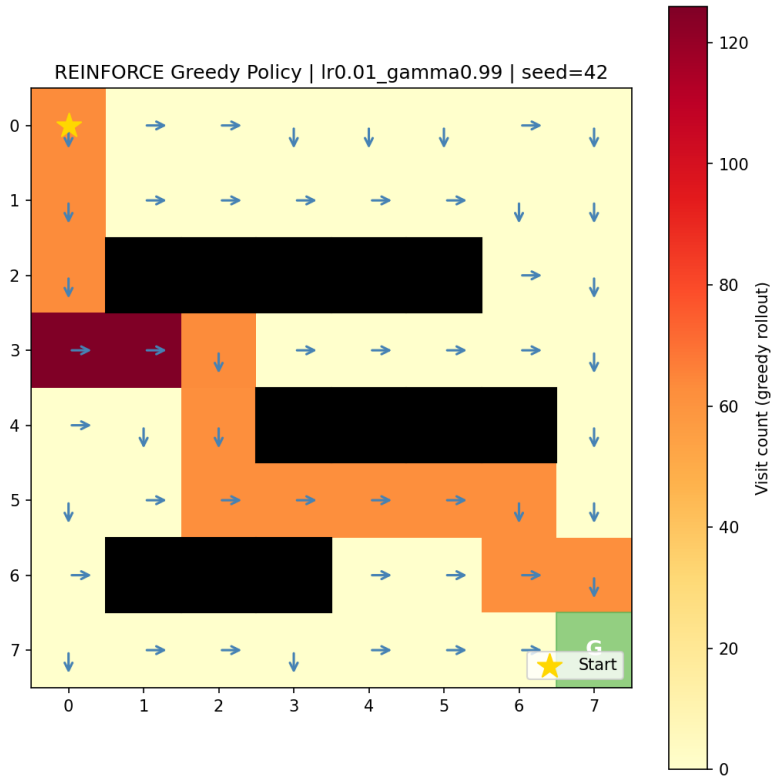


**Figure 7:** REINFORCE greedy policy (best config, seed 42): Viable but more exploratory. Scattered visitation reflects suboptimal actions and the limitations of on-policy learning.

Both policies solve the maze, but DQN's experience replay enables sharper value estimates and a cleaner greedy policy.

# 7 Key Insights

## 7.1 Variance and Stability

The 20-fold variance reduction in DQN relative to REINFORCE is practically significant. Experience replay decouples updates from trajectory quality by maintaining a buffer of 5 000 diverse past experiences. REINFORCE must exploit current trajectories immediately; poor early episodes can trap the policy in local optima.

## 7.2 Why REINFORCE Exhibits High Variance

1. **Gradient noise:** Monte Carlo returns are unbiased but high-variance. Small trajectory differences can flip the gradient direction.
2. **On-policy sampling:** Cannot reuse past experience. Early failures accumulate as weak gradient signals.
3. **Sparse rewards:** Approximately 90% of episodes reach no goal; the signal-to-noise ratio is very low.

Potential mitigations include baseline subtraction (Actor-Critic), entropy regularisation, or larger batch sizes.

### 7.3 Hyperparameter Sensitivity

**DQN:** Optimal at $\alpha = 0.001$. Higher rates cause Q-value overestimation and oscillatory instability. Once tuned, performance is stable across seeds.

**REINFORCE:** Requires a mid-range $\alpha = 0.01$. Too low (0.001) gives slow learning; too high (0.1) amplifies gradient noise. On-policy methods are more fragile at hyperparameter extremes.

Both algorithms critically depend on $\gamma = 0.99$ for multi-step planning; $\gamma = 0.9$ proves too myopic for the 15-step maze horizon.

### 7.4 Environment Design Considerations

This configuration (sparse rewards, stochastic transitions, long horizon) inherently favours DQN. Under dense rewards, deterministic dynamics, or shorter paths, conclusions would likely differ. Algorithm comparisons are environment-dependent; problem design should be chosen to match research questions.

## 8 Conclusion

Both DQN and REINFORCE successfully learned the maze task, each achieving a 93% improvement over the random baseline. DQN dominates on stability and convergence speed; REINFORCE shows comparable mean performance with variance that could be substantially reduced through established techniques. Key findings are summarised below:

1. Experience replay reduces variance by 20× — a critical stabiliser for value-based methods in sparse-reward settings.
2. Hyperparameter tuning is essential; both algorithms fail at inappropriate learning rates.
3. Mean performance alone masks variance differences; comprehensive error reporting is required for fair comparison.
4. Environment design directly influences algorithm comparison outcomes.
5. JAX's functional paradigm enabled an efficient, clean implementation with minimal boilerplate.

Future work could explore Actor-Critic architectures, TRPO/PPO, or Gymnax (JAX-native environments) for further improvements.